Lecture 01 — Why linear algebra? Vectors as representations

Mon Aug 24

Last updated: August 20, 2026.

Reading: Boyd 1.1. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Identify objects that can be represented as vectors.
2. Interpret vector entries, units, and indexing in context.
3. Distinguish scalars from vectors.
4. Translate a small data, position, signal, or measurement problem into vector form.

Warm-up

For each description, decide: could this naturally be written as a vector? If yes, guess a sensible n and what the entries would mean. If no, say why not.

1. Today's high temperature in Arcata, in °F.
2. High temperatures in Arcata on Mon–Thu this week, in °F.
3. The phrase "mostly foggy."
4. An RGB screen color with red, green, and blue intensities between 0 and 1.

Fill in

An *n-vector* is an ordered list of ______________ numbers, written

$$x = (x_1, x_2, \dots, x_n) \quad \text{or} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

The number n is the ______________ (or dimension) of x. Each x_i is an ______________ of x. In this course we index from $i = 1$ to $i = n$ (Boyd's convention).

Two n-vectors a and b are equal when ___________________.

A *scalar* is ___________________. The number x_3 is a ____________; the list x is a ____________.

The *zero vector* 0 has ______________. The *ones vector* $\mathbf{1}$ has ______________. The *standard unit vector* e_i has _____________________.

In an application, we often keep a separate list of _____________ so that we know what each entry means and what ___________ it has. Entries with different units are still allowed in one vector; comparing them as “the same kind of quantity” is _____________.

Worked in class

W1. An Arcata weather station records four successive high temperatures, in °F:

$$t = (63, 59, 61, 66).$$

What is n? What is t_3, and what does it mean in context? Is 63 a vector or a scalar? Could you add a fifth day without changing the meaning of $t_1, \ldots, t_4$?

W2. A redwood seedling is described by height in cm, basal diameter in mm, and soil moisture in percent. Write a 3-vector f for a seedling that is 18 cm tall, 4 mm across, and 27% moisture. Give an entry label for each index. Which entries are allowed to be compared as “the same kind of quantity,” and which are not?

Your turn

Y1. An RGB color is the 3-vector $c = (0.15, 0.80, 0.35)$, with entries between 0 and 1 for red, green, and blue. Which entry is largest, and what color does that suggest? What 3-vector is black in this encoding?

Y2. A short field note is scored against the dictionary (fog, rain, wind). The word-count vector is $w = (7, 2, 0)$. What is n? What does w_1 mean? What does $w_3 = 0$ tell you?

Y3. A trail junction is 120 m east and 40 m north of a trailhead. Write a 2-vector p for that location in (east, north) meters. What would $(-20, 10)$ mean as a second location in the same encoding?

Answers (Your turn)

Y1. The second entry is largest, so the color is mostly green. Black is $(0, 0, 0)$.

Y2. $n = 3$. The word *fog* appears 7 times. *Wind* does not appear.

Y3. $p = (120, 40)$. The point $(-20, 10)$ is 20 m west and 10 m north of the trailhead.